LATANOPROST

Once daily at night

Iris colour change

CHEMICAL NAME Isopropyl (Z)-7-[(1R,2R,3R,5S)-3,5-dihydroxy-2-[(3R)-3-hydroxy-5-phenylpentyl]cyclopentyl]hept-5-enoate

CLASS Prostaglandin F2α analogue (ophthalmic antiglaucoma)

TRADE NAMES Xalatan · Monopost (preservative-free) · Xalacom (with timolol)

THERAPEUTIC USES / INDICATIONS

- Open-angle glaucoma and ocular hypertension — often first-line.
- Paediatric glaucoma.

ADMINISTRATION

- **Eye drop** once daily in the evening (more often reduces effect).
- Remove contact lenses; reinsert after 15 min.
- Wait 5 min between other eye drops.
- Store unopened bottle in fridge (opened at room temperature up to 4 weeks).

DOSAGES

- 0.005%: 1 drop once daily at night in affected eye(s).

PHARMACOKINETICS

- **Onset:** 3–4 h; max effect 8–12 h. **Duration:** ≥24 h.
- Prodrug hydrolysed in cornea to active acid.
- Systemic metabolism in liver; minimal systemic effects.

NURSING MANAGEMENT

- Teach drop technique; once daily in the evening.
- Warn about iris colour and eyelash changes (especially if treating one eye).
- Monitor IOP.

PHARMACODYNAMICS

- Agonist at **FP prostaglandin receptors** → ↑ uveoscleral outflow of aqueous humour → ↓ IOP (25–35%).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Eye redness (conjunctival hyperaemia), stinging.
- Darkening of iris (brown pigmentation; permanent) especially mixed-colour irises.
- Eyelash growth (longer, thicker, darker); periorbital skin darkening.

ADVERSE EFFECTS (SERIOUS)

- Cystoid macular oedema (aphakic/pseudophakic patients).
- Iritis/uveitis; reactivation of herpes keratitis.
- Periorbital fat atrophy (prostaglandin-associated periorbitopathy).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** active uveitis, history of herpes keratitis, aphakia, pregnancy.

KEY INTERACTIONS

- **Other prostaglandin analogues:** paradoxical IOP rise; do not combine.
- **Thiomersal-containing drops:** precipitate — separate by 5 min.

PATIENT EDUCATION

- Use one drop every evening — more often makes it work less well.
- Eye colour may slowly become browner and eyelashes longer; this is permanent for eye colour.
- Wipe excess from eyelid skin to reduce skin darkening.
- Report eye pain or vision changes.

KEY POINTS

- Prostaglandin analogue — ↑ uveoscleral outflow.
- Iris darkening, eyelash growth, red eye.
- Once daily at night.
- First-line for open-angle glaucoma.

SOURCES CHECKED [980] Latanoprost / Timolol 50 micrograms / ml + 5 mg / ml Eye Drops, Solution (UK SmPC)

Full references p. 432–457